Section 1: The Problem Definition (Domain and Theme)

For this project we have chosen to make a cyclical cellular automaton, using the stencil pattern. Specifically, we will be implementing both neighboring strategies, the von Neumann neighborhood and the Moore neighborhood.

Section 2: The Data Strategy (Input & Output)

We will obtain our datasets by either manually creating them, or by running a noise generation algorithm that we will implement.

Input Generation Algorithms

- Random Noise Generation: each point will have a randomly generated value.
- Sparse Seed Generation: each point will be set to 0, then we will place N amount of points into the seed of a high value.
- Random Noise Shape Generation: We first generate a set of points by running the Random Noise Generation Algorithm, and then we place N amounts of shapes at random locations where each point's value increases to the maximum possible value gradually towards the shape's core.

Input Sizes

- Small: (512 x 512) For debugging and validity testing.
- Medium: (1920x1200) Demo show size.
- Large: (16384x16384): Stress test for all 3 variants.

Section 3: Goals & Verification Plan

Output: This program will generate a bmp file that will get aggregated into an mp4 file through ffmpeg.
Verification: To test the validity of the code, we will perform a checksum of the final bmp of the serial and parallel implementation for the same seed, if the outputs then we'll know that one of our algorithms is incorrectly implemented.
Performance: We aim to achieve a near linear speedup when using OpenMP, diminishing returns for the MPI implementation due to the overhead of IPC.